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Mutations in two component system (PhoPQ and PmrAB) in colistin resistant *Klebsiella pneumoniae* from North Indian tertiary care hospital

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Abstract

Colistin resistance in Gram negative bacteria is mainly attributed to chromosomal mutations in Two Component Systems (TCS) PhoPQ and PmrAB and plasmid-borne genes (*mcr* and its variants). The aim of this study was to understand the molecular basis of colistin resistance in *Klebsiella pneumoniae* and determine clonal transmission, in a North Indian tertiary care hospital over a 2.5 year period. Antimicrobial susceptibility was determined by Vitek and colistin resistance was confirmed by broth microdilution. Carbapenemases (*bla*_{KPC}, *bla*_{VIM}, *bla*_{IMP}, *bla*_{NDM}, *bla*_{OXA-48}) and *mcr*-1 screening was done by PCR. Mutations in chromosomal genes *mgrB*, *phoP*, *phoQ*, *pmrA*, *pmrB* were analysed. Sequence typing was performed by Multilocus sequence typing (MLST). OXA-48 was detected in thirteen isolates while three isolates co-expressed OXA-48 and NDM. The *mcr*-1 gene was absent in all 16 isolates. Deleterious mutations in *mgrB* included insertion sequences IS903 and ISkpn26 and a premature stop codon. A total of 18 point mutations were identified in PhoPQ and PmrAB TCS; of which, novel mutations were reported in *phoQ* (K46E, L322V, D152N, F373L, R249G), *pmrB* (P159R) and *pmrA* (D149L). Six different sequence types ST231, ST147, ST395, ST42, ST14 and ST101 were identified. Phylogenetic analysis showed that sequence types ST14, ST395 and ST147 are closely related to ST101 and all identified sequence types had a common ancestor ST231. Colistin resistance in *K. pneumoniae* was attributed to mutations in PhoPQ and PmrAB TCS, while location specific distribution of strains indicates clonal transmission. The results of this study will help in formulation of effective infection prevention and antimicrobial development strategies.

Introduction

Colistin isolated from *Paenibacillus polymyxa* in 1947 is a polycationic antibiotic active against Gram negative bacteria [1]. After its use was restricted in 1980s due to neurotoxicity and nephrotoxicity, colistin use increased in the early 2000s with increasing numbers of multidrug resistant organisms (MDR) [2]. The injudicious use of colistin in human, animal and environment has led to the emergence of

resistance. Colistin interacts with the phosphate groups of lipid A region of lipopolysaccharides (LPS) [1]. Once attached, it damages the outer membrane (OM) by displacing divalent cations Ca²⁺ and Mg²⁺ causing cell death. Loss of cations in the OM activate the Two Component System (TCS) PhoPQ and PmrAB which results in the synthesis and transfer of phosphoethanolamine (pEtN) and 4-amino-4-deoxy-L-arabinose (L-Ara4N) cationic groups. These LPS modifications neutralise negative charge on outer membrane and hinder colistin binding [3].

Mutations in the genes of PhoPQ and PmrAB TCS result in its constitutive activation and results in colistin resistance [1, 3]. Colistin resistance could also result by the acquisition of plasmid-based gene *mcr* and its variants leading to addition of pEtN to lipid A [4]. Besides, a variety of mechanisms like efflux pumps, capsule formation and over-expression of the outer membrane protein OprH have also been reported in colistin resistant Gram negative bacteria [3].

Among all the Gram negative bacteria, *Klebsiella pneumoniae* has the highest rate of resistance of 2–8%

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towards colistin in India [5]. The present study was carried out in a North Indian tertiary care hospital where ~15% patients are of foreign origin. In addition, as per the hospital antibiogram colistin resistance in *K. pneumoniae* is reported to be 1–3% in isolates from intensive care units. Recently, few studies have reported the molecular characterisation of colistin resistance in isolates from South India [6]. However, there are only few studies reported from Northern India. In view of this, we have performed the molecular characterisation of colistin resistance in *K. pneumoniae* clinical isolates to identify the genetic basis of resistance and understand the clonal relationship between the prevalent strains.

Methods

Bacterial strains

This prospective experimental study was carried out in a 450 bed tertiary care hospital in Gurugram, Haryana, India. All clinical samples that were received in the microbiology department during the study period (January 2017–June 2019) were processed as per the routine microbiological testing protocols. Bacterial species identification and antibiotic susceptibility testing was done using VITEK 2 Compact system (bioMérieux). Antibiotics tested for susceptibility included broad groups cephalosporins, carbapenems, colistin, piperacillin-tazobactam and trimethoprim-sulfamethoxazole. Minimum Inhibitory Concentration (MIC) was interpreted as per Clinical and Laboratory Standards Institute (CLSI) guidelines [7]. 15 out of 1689 *K. pneumoniae* clinical isolates resistant to colistin (MICs > 2 mg l⁻¹) and one multidrug resistant *K. pneumoniae* susceptible to colistin (MIC = 2 mg l⁻¹) as per standard laboratory practice for determining resistance were included in the study.

Broth microdilution (BMD) and E-test

Colistin MIC was determined for all 16 isolates using broth-microdilution (MIKROLATEST MIC Colistin, Erba Transasia) and E-test (Colistin Ezy MICTM Strip, HiMedia) as per the manufacturers' protocols. Known mcr-1 positive strain was used as positive control and *Pseudomonas aeruginosa* ATCC 27853 and *Escherichia coli* ATCC 25922 were used as negative controls. MIC was interpreted as per European Committee on Antimicrobial Susceptibility Testing (EUCAST) guidelines [8]. The variation in MIC by the three different methods [Vitek, BMD (broth microdilution) and E-test] was statistically evaluated keeping BMD as the gold standard. For data analysis purposes, MIC of

≥ 16 mg l⁻¹ determined by Vitek was considered as 16 mg l⁻¹. For all statistical tests, SPSS (21.0.0) was used and *p* value < 0.05 indicated statistical significance of the data.

Screening for carbapenemases

DNA extraction was performed on single colony isolates from the culture plate as described previously [9]. PCR was performed for screening of Carbapenemases *bla*_{KPC}, *bla*_{VIM}, *bla*_{IMP}, *bla*_{NDM} and *bla*_{OXA-48} in all selected *K. pneumoniae* isolates using primers and conditions described previously [9, 10].

Molecular characterisation of colistin resistance

PCR amplification of chromosomal genes *phoP*, *phoQ*, *pmrA*, *pmrB* and *mcrB* was performed using primers and PCR conditions as described earlier [11] and PCR for plasmid based mcr-1 gene was performed as per Liu et al. [4]. Known mcr-1 positive strain was used as positive control. The PCR products were sequenced by Sanger dideoxy sequencing method using ABI, Genetic Analyzer 3500xL.

The sequences obtained for the two component system genes (*phoP*, *phoQ*, *pmrA*, *pmrB* and *mcrB*) were analysed for presence of any mutations using known *K. pneumoniae* sequences from NCBI GenBank and aligning the sequences using BLAST software of NCBI (<http://blast.ncbi.nlm.nih.gov>). The effect of mutation as neutral/ deleterious was determined by freely available software PROVEAN (Protein Variation Effect Analyzer) v1.1.3 (<http://provean.jcvi.org/index.php>). 16 *mcrB* gene sequences were submitted in NCBI GenBank and have been assigned the accession IDs MK479286–MK479294, MN307741 and MN187243–MN187248.

Sequence typing by MLST

Multilocus sequence typing (MLST) was performed to identify the sequence type using seven housekeeping genes, primer sequences and PCR conditions as described by Diancourt et al. [12]. Allele number on each housekeeping gene was assigned and sequence type was identified by matching the allele numbers to the profile database as per protocol available on the Institute of Pasteur website (<http://bigsd.bpasteur.fr/klebsiella>).

Phylogenetic analysis

Phylogenetic relationship between sequences was identified using CLUSTAL W2 Package (www.ebi.ac.uk/Tools/phylogeny/simple_phylogeny).

Results

Epidemiology

In the entire study period, 8208 bacterial isolates were processed comprising of 6765 Gram negative organisms. Among the Gram negative bacteria 2928 were *E. coli*, 1689 *Klebsiella* sp., 489 *Acinetobacter* sp. and 762 *Pseudomonas* sp. As per Vitek results, colistin resistance was reported in 15 *K. pneumoniae*, 1 *E. coli* and 1 *Acinetobacter baumannii*. Since Colistin Resistant *K. pneumoniae* (ColR-KP) showed predominance, we selected it for further analysis.

The MDR strains of *K. pneumoniae* in the present study were isolated from various clinical samples of 14 in-patients from the medical surgical Intensive care unit (ICU) (11) [ICU-1 (5) and ICU-2 (6)], neurology ICU (1), bone marrow transplant unit (1) and ward (1). Patients belonged to different geographical regions such as Uzbekistan (1), Iraq (1), Kenya (1), Nigeria (1) and India (10) (Haryana-8, Delhi-1, Punjab-1).

Of the 16 *K. pneumoniae* strains selected, colistin resistance was confirmed in 13 (81.25%) by BMD and only 8(50%) by E-test. Considering BMD as the gold standard for confirmation of colistin MIC as jointly recommended by CLSI and EUCAST [13] we observed highest categorical agreement between BMD and Vitek2 (45%) with 13.3% MEs (major errors denoting false resistant results) and no VMEs (very major errors denoting false susceptible results) Antibigram and MIC determination for these isolates is presented in Table 1.

Molecular mechanisms of colistin resistance

Of the 16 isolates, thirteen harboured only *bla_{OXA-48}* gene while three co-expressed *bla_{NDM}* and *bla_{OXA-48}*. Other carbapenemases *bla_{KPC}*, *bla_{VIM}* and *bla_{IMP}* were absent in our study. Isolates producing *bla_{NDM}* alone were not detected (Table 2).

Plasmid mediated resistance due to *mcr-1* was not detected in any of the isolates. Molecular analysis of chromosomal genes of TCS PhoPQ and PmrAB showed point mutations. More than one point mutation was found in a few isolates which altogether added to 18 different mutations (Table 2). A few mutations were found in more than one isolate taking the total number to 29. Analysis of point mutations in PROVEAN revealed that out of 18, 7 were deleterious and rest (11 mutations) were neutral. All *phoP* gene sequences were found to be wild type. In *phoQ* gene six mutations were reported, out of which three mutations K46E, L322V and G385C were deleterious while the other three mutations R249G, F373L and D152N were silent. Among *pmrA/B*, five mutations were observed in *pmrA* and seven in *pmrB* genes. In *pmrA* gene one mutation

G53V was deleterious whereas four others N131D, G144D, D149L and N219H were silent. Likewise, in *pmrB* gene three mutations R256G, P159 R and T157P were found to be deleterious and four mutations L213M, A246T, A228T and A5V reported to be silent. Of all these mutations, seven mutations (five in *phoQ* and two in *pmrA*) are being reported for the first time in this study.

Mutations in *mgrB* were found in three out of sixteen isolates. Insertional inactivations were detected in two isolates, IS903-like mobile element at nt 77 of *mgrB* gene (GenBank accession no. MK479294) and ISKpn26-like element at nt 75 (GenBank accession no. MN307741). Nonsense mutation c88t which caused premature termination of *mgrB* was found in one isolate (GenBank accession no. MK479286).

MLST and phylogenetic analysis

The sixteen isolates showed six different sequence types that included ST231 ($n=6$), ST147 ($n=5$) and ST395 ($n=2$), ST42 ($n=1$), ST14 ($n=1$) and ST101 ($n=1$). Five of the six ST231 were isolated from ICU-2 while majority (4/5) of ST147 was from ICU-1. There was no association between similar mutations with a particular sequence type. The phylogenetic tree was constructed based on concatenated sequences of seven housekeeping genes used for MLST. Isolates of sequences ST14, ST395 and ST147 were closely related to ST 101. It was also clear that ST231 is the common ancestor for all the sequence types reported in the study (Fig. 1).

Discussion

Antimicrobial resistance (AMR) is a serious public health threat worldwide. Gram-negative bacteria in particular, have exhibited the emergence of multi-drug resistance (MDR), extensive drug resistance (XDR) and further difficult-to-treat resistance [14]. In India, the situation of AMR is even worse with the rates of pan drug resistance ranging from 3.5 to 8.1%, posing serious therapeutic challenge [5, 15]. In the present study, all the 16 isolates showed resistance to first-line agents, including β -lactams, β -lactamase inhibitor combinations, fluoroquinolones and aminoglycosides. Colistin resistance in *K. pneumoniae* in our hospital setting was relatively low compared with other hospital centres in India and was around 0.9% (15/1689) in this study. The low prevalence could also be the result of using Vitek as a screening procedure which is not a very sensitive tool to detect colistin resistance. However, identification of mutations in colistin resistant strains and routes of transmission at this initial stage is crucial in formulating appropriate antimicrobial stewardship strategies.

Table 1 Antibigram and Colistin minimum inhibitory concentration (MIC) determination of *Klebsiella pneumoniae*

S. No	Strain	Source	MIC (mg l ⁻¹)													
			BMD*		E-test	Vitek	Colistin interpretation	Cephalosporins (FEP, CFM, CAZ, CRO)			AMK	GEN	AMC	CIP	IPM	MEM
1	ARTLCOL1	Blood	8	4	4	>16	R	64	64	16	32	4	8	16	128	320
2	ARTLCOL2	sputum	2	1.5	2	>16	S	64	64	16	32	4	4	16	128	320
3	ARTLCOL3	Urine	8	4	4	>16	R	64	64	16	32	4	16	16	128	320
4	ARTLCOL4	Blood	8	3	3	>16	R	64	64	16	32	4	16	16	128	320
5	ARTLCOL5	Pus	16	2	2	>16	R	64	64	16	32	4	16	16	128	320
6	ARTLCOL6	Urine	0.5	1.5	1.5	>16	S	64	64	16	32	4	16	16	128	320
7	ARTLCOL7	Drain Fluid	0.5	1.5	1.5	>16	S	64	64	16	32	4	16	16	128	320
8	ARTLCOL8	Blood	16	8	8	>16	R	64	64	16	32	4	4	8	128	320
9	ARTLCOL9	Pus	16	8	8	>16	R	64	64	16	32	4	16	16	128	320
10	ARTLCOL10	Urine	16	8	8	>16	R	64	64	16	32	4	16	16	128	320
11	ARTLCOL11	Pus	16	16	16	>16	R	64	64	16	32	4	4	16	128	320
12	ARTLCOL12	Blood	16	2	2	>16	R	64	64	16	64	4	8	16	128	320
13	ARTLCOL13	Urine	4	2	2	4	R	64	64	16	32	4	4	16	128	40
14	ARTLCOL15	ET	8	1.5	1.5	>16	R	64	64	16	32	4	16	16	128	320
15	ARTLCOL16	Urine	4	1.5	1.5	4	R	64	64	16	32	4	16	16	128	320
16	ARTLCOL17	ET	16	16	16	>16	R	64	64	16	32	4	16	16	128	320

Degree of agreement

Comparison of MIC determination methods

	BMD ^a		E-test		Vitek	
Susceptible isolates (<i>n</i>)	3		8		1	
Resistant isolates (<i>n</i>)	13		8		15	
Accuracy			69%		88%	
Categorical Agreement (CA)			0.38 (p=0.04)		0.45 (0.03)	
Very Major Errors (VME) ^b			5 (62.5%)		0	
Major Errors (ME) ^c			0		2 (13.3%)	
Sensitivity			62%		100%	
Specificity			100%		33%	
Positive Predictive Value (PPV)			100%		87%	
Negative Predictive Value (NPV)			38%		100%	

AMK amikacin, AMC amoxicillin/clavulanic acid, FEP cefipime, CFM cefixime, CAZ ceftazidime, CRO Ceftriaxone, CIP ciprofloxacin, GEN gentamicin, IPM imipenem, MEM meropenem, TAZ piperacillin/tazobactam, TMP trimethoprim/sulfamethoxazole

^aBMD, Broth microdilution is considered standard method for MIC determination. MIC of Vitek and E-test was compared with BMD to statistically evaluate the correlation between the three methods

^bVMEs (very major errors denoting false susceptible results)

^cMEs (major errors denoting false resistant results)

Table 2 Mutation analysis of two component system (PhoPQ and PmrAB) in *K. pneumoniae* isolates

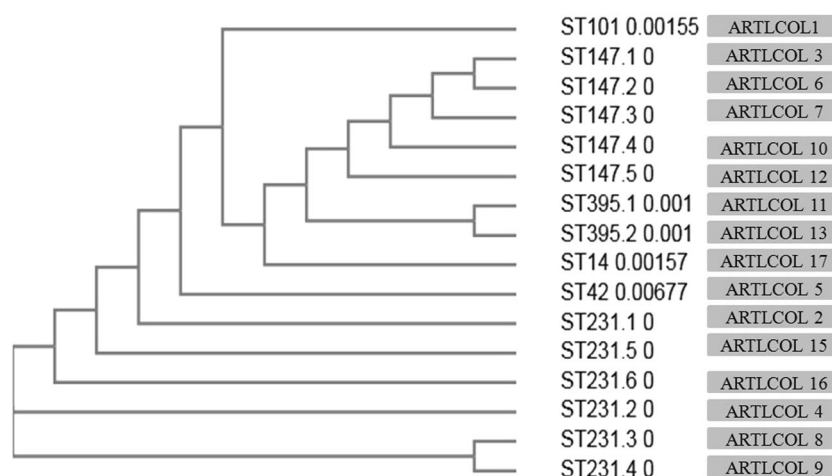
Strain	Sequence type	Location	Nationality of patient	BMD MIC (mg/L)	Mutations in genes ^a				Carbapenemase
					<i>mgrB</i>	<i>phoP</i>	<i>phoQ</i>	<i>pmrA</i>	<i>pmrB</i>
ARTLCOL1	ST101	5th Floor	Uzbekistan	8	C88T (Q30Stop)	WT	WT	WT	L213M-Neutral
ARTLCOL2	ST231	2nd ICU	Gurugram(India)	2	T134G (F45S) Neutral	WT	WT	WT	L213M-Neutral
ARTLCOL3	ST 147	2nd ICU	Gurugram(India)	8	WT	WT	D152N-Neutral	WT	R256G- Deleterious, A246T Neutral
ARTLCOL4	ST231	1st ICU	Iraq	8	WT	WT	WT	G53V Deleterious	A246T- Neutral
ARTLCOL5	ST 42	4th ICU	Gurugram(India)	16	IS903	WT	WT	D131N, G144D, D149L, N219H-Neutral	A228T- Neutral, A5V- Neutral
ARTLCOL6	ST 147	1st ICU	Rewari(India)	0.5	WT	WT	R249G - Neutral	WT	WT
ARTLCOL7	ST147	1st ICU	Rewari(India)	0.5	WT	WT	F373L - Neutral	WT	WT
ARTLCOL8	ST231	2nd ICU	Nigeria	16	WT	WT	K46E, L322V Deleterious	WT	WT
ARTLCOL9	ST231	2nd ICU	Nigeria	16	WT	WT	K46E, L322V Deleterious	WT	WT
ARTLCOL10	ST147	1st ICU	Delhi(India)	16	WT	WT	WT	WT	P159 R, R256G Deleterious A246T-Neutral
ARTLCOL11	ST395	5th Floor	Kenya	16	ISKPN26	WT	WT	WT	WT
ARTLCOL12	ST147	1st ICU	Gurugram(India)	16	WT	WT	WT	G53V Deleterious	R256G Deleterious, A246T- Neutral
ARTLCOL13	ST395	Neurology ICU	Rohtak(India)	4	WT	WT	WT	WT	T157P- Deleterious
ARTLCOL15	ST231	2nd ICU	Rewari(India)	8	WT	WT	WT	WT	R256G Deleterious
ARTLCOL16	ST231	2nd ICU	Mahendargarh (India)	4	WT	WT	WT	WT	R256G Deleterious
ARTLCOL17	ST14	2nd ICU	Punjab(India)	16	WT	WT	G385C-Deleterious	WT	WT

• Novel Missense mutations- *phoQ* (K46E, L322V), *pmrB* (P159R)• Novel Silent mutations- *phoQ* (D152N, F373L, R249G), *pmrA* (D149L)• Silent mutations- *pmrB* (L213M, A246T, A5V, A228T), *pmrA* (G144D, D131N, N219H)

• WT wild type

^aDeleterious or neutral effect of mutations has been checked using PROVEAN software

Fig. 1 Phylogenetic analysis and clonal relatedness of *K. pneumoniae* sequences. Phylogram showed that sequence types ST395 and ST147 are closely related to ST101 and all the identified sequence types have a common ancestor ST231. ST101 and ST231 are hypervirulent sequence types reported globally



VITEK 2 Compact system, a commonly used clinical microbiology analyser does not determine colistin resistance reliably [16]. Therefore, BMD and E-test were used for further confirmation of MIC. The agreement between BMD and Vitek (81.3%) was higher than E-test (50%) for determining colistin resistance which is similar to previous reports [17]. Disk diffusion and E-test are not recommended for MIC determination of polymyxins as large cationic peptide molecules do not diffuse through agar well and give discrepant results [18].

Cross-resistance refers to acquisition of resistance to a particular drug which causes simultaneous resistance to other drugs as well. Antibiotic resistant strains generally exhibit significant changes in the MICs to more than one antibiotic [19]. This probably explains the high MIC of colistin obtained using Vitek in two isolates which were sensitive as per BMD. Mutational resistance and horizontal gene transfer contribute significantly in developing co-existence of resistance genes in bacteria. Molecular characterisation of three ColS-KP isolates showed presence of carbapenem-resistance genes. However, genes of PhoPQ and PmrAB TCS had either wild type genes or silent mutations explaining their susceptibility to colistin in these isolates.

Multiple point mutations were detected in *PhoPQ* and *PmrAB* TCS, including missense and silent mutations. In this study we found both novel and already known mutations. All the mutations reported in *phoQ* gene were novel except for one G385C which is deleterious in nature [20]. On the contrary, only one novel mutation (D149L- silent) was reported in *pmrA*, while the others were already known [3, 20–22]. In *pmrB*, T157P (deleterious) was known mutation [3, 11] whereas R256G, and P159R were novel missense mutations. Rest of the silent mutations (L213M, A2246T, A228T and A5V) in *pmrB* were reported in earlier studies [3, 20, 23, 24]. Overall, our study showed the presence of at least one deleterious mutation in the resistant strain. Association of increased MIC with presence of

multiple deleterious mutations could not be established in this study as our BMD kit could detect susceptibility only a maximum of $16 \mu\text{g ml}^{-1}$.

Mutation associated with *mgrB* is the commonest mechanism that leads to upregulation of *phoPQ* two component system and results in modification of LPS [23]. These mutations include insertional inactivation and point mutations in *mgrB*. In our study we reported disruption of *mgrB* due to IS903-like and ISKpn26-like insertion elements. This is in concordance with earlier studies [23, 24]. Premature termination of *mgrB* at the 30th amino acid concurs with international publications but to the best of our knowledge this is the first report from India [24, 25]. Incidentally, the *K. pneumoniae* strain harbouring this mutation was isolated from a patient of Uzbek origin. This observation underlines the importance of stricter surveillance in the era of medical tourism.

MLST reported diverse sequence types among which ST231, ST147, ST395 and ST14 were earlier reported in carbapenem-resistant *K. pneumoniae* from India [26]. Furthermore, ST14 and ST147 have been reported as XDR *K. pneumoniae* high risk clones internationally [27]. ST101 carrying *bla_{KPC}* is reported as an international epidemic clone worldwide [28]. However, the ST101 sequence type in our study harboured OXA-48 and had a truncated *mgrB*, which is probably a first of its kind observation. Sequence type ST42 which harboured the insertion sequence IS903-like in *mgrB* gene in our study has previously been found to carry mcr-8 plasmid [29]. This shows that exposure to colistin might lead to resistance by acquisition of different mutations.

The distribution of same sequence types in isolates from the same location indicates the clonal expansion and transmission of MDR *K. pneumoniae* isolates within the hospital setting. The predominance of sequence types ST231 and ST147 in ICU-2 and ICU-1 respectively supports this observation although a confirmation of the same

can only be achieved by whole genome sequencing. This information can be used in improving our surveillance and infection control programs.

Phylogram suggested that ST231 is the common ancestor for all the sequence types reported. ST147, ST14 and ST395 were closely related to ST101. In recent studies, ST101 lineage has reported 11% increase in mortality rate in comparison to non-ST101 infections. ST101 was the predominant lineage found to be associated with the dissemination of carbapenemases by IncL plasmids carrying blaOXA-48/bla_{OXA}-181 genes [30]. Out of the six (43%) fatal cases in our study, three were associated with ST231, two with ST147 and one with ST395.

The findings of the present study are to the best of our knowledge, one of the early reports on molecular characterisation of colistin resistance in *K. pneumoniae* from northern India. Another salient feature of our study is that it was conducted in a hospital which receives international patients that gives us an idea of the global transmission of resistance genes and virulent strains. The limitation of the study is that we could not elucidate in detail the exact effect of individual mutations using gene expression studies and complementation assays. Moreover, whole genome sequencing would have given a detailed pattern of resistance genes and virulence factors.

In conclusion, the present study reported mutational analysis of colistin resistant *K. pneumoniae* strains in a North Indian tertiary care hospital. We observed that mutations in PhoPQ and PmrAB TCS cause colistin resistance and isolation of similar sequence types confirm the clonal transmission of ColR-KP within the hospital setting. This information is vital in designing effective infection prevention strategies. Furthermore, understanding the evolution of resistant strains may help in formulating appropriate antimicrobial stewardship strategies.

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Author contributions All authors contributed to the study conception and design. PKN—Study design, Laboratory methods, data analysis and manuscript writing; NC—Study design, Laboratory methods; manuscript correction RP—Isolate collection and Antibiotic susceptibility testing; MD and NJ—Study design and manuscript correction. All authors read and approved the final manuscript.

Compliance with ethical standards

Conflict of interest The authors declare no competing interests.

Ethics statement and consent for publication The study protocol was approved by Artemis Hospital Institutional Ethics Committee (AHS-IEC Reg No: ECR/53/Inst/HR/2013/RR-16). Waiver for consent was obtained as microbiological samples were considered a part of standard care procedures and keeping patient identity anonymous in the study. All authors provide their consent for publication.

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